

Residue Manifold Learning: Constraint Sampling, Structural Recovery, and Phase-Lock Geometry

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Abstract

We study structure, sampling, and representation in modular arithmetic through the mod30 residue manifold. Prime-support residues form an eight-lane discrete manifold embedded in $\mathbb{Z}/30\mathbb{Z}$. We show that constraint-aligned sampling improves signal access, that nonnegative matrix factorization (NMF) recovers the manifold compactly, and that sparse autoencoder (SAE) configurations can fragment or dilute structure despite comparable reconstruction performance. We introduce the Constraint-Guided Coherence Score (CGCS), a bounded metric for structural fidelity, and demonstrate that representation regimes separate into phase-locked, partial, and diluted regions. Finally, we give a geometric interpretation of CGCS as a cosine alignment constraint with a phase-lock boundary near 45° , unifying sampling, representation, and geometry.

1 Introduction

Structure, sampling, and representation interact in determining whether underlying patterns can be recovered from data. Recent work has emphasized that access to structured representations can eliminate apparent computational advantages.

We investigate this interaction in a controlled modular setting. Prime residues modulo 30 form a discrete, structured manifold, providing a minimal environment to study:

- Structure: existence of constrained residue lanes
- Sampling: access aligned vs unaligned with structure
- Representation: recovery vs fragmentation of structure

Our goal is to isolate how these components combine to determine structural fidelity.

2 Residue Manifold Structure

Let $n \in \mathbb{Z}$ and consider residues modulo 30:

$$r = n \bmod 30.$$

Prime numbers lie only in:

$$\{1, 7, 11, 13, 17, 19, 23, 29\}.$$

These residues define an eight-lane discrete manifold embedded in $\mathbb{Z}/30\mathbb{Z}$.

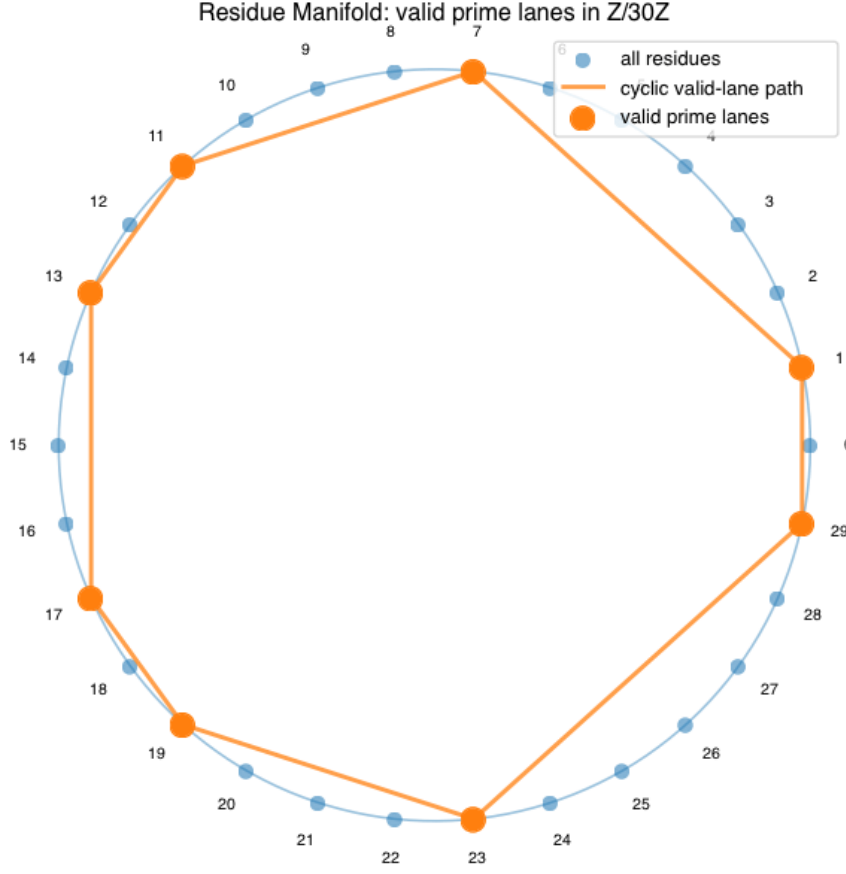


Figure 1: **Residue manifold structure.** Valid prime-support residues form eight discrete angular lanes, revealing a structured subset within $\mathbb{Z}/30\mathbb{Z}$.

This structure exists independently of any representation method.

3 Constraint Sampling

We compare:

- Uniform sampling over all residues
- Sampling restricted to valid lanes

Uniform sampling yields:

$$\text{signal rate} \approx \frac{8}{30},$$

while constrained sampling achieves:

signal rate = 1.

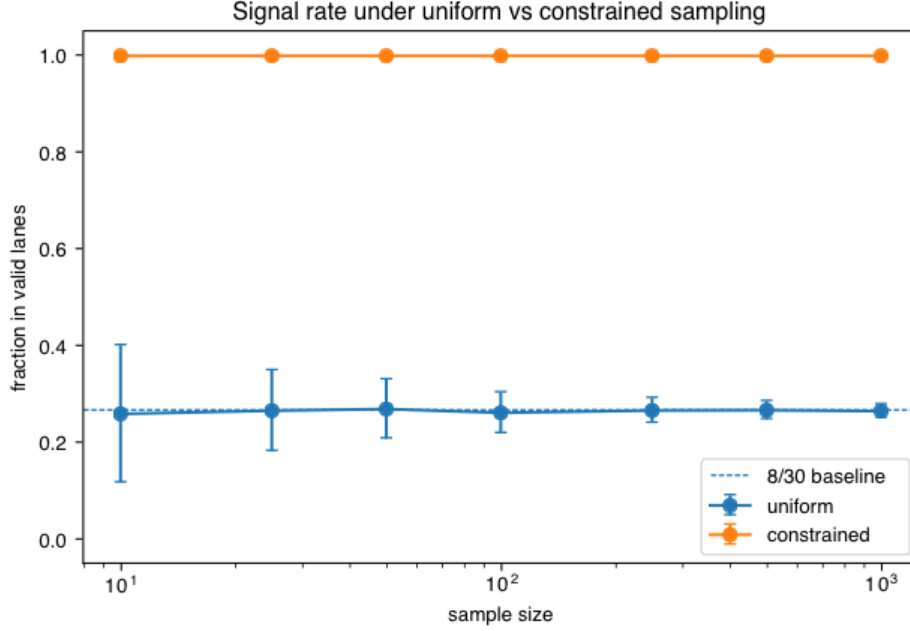


Figure 2: **Sampling efficiency.** Uniform sampling spreads probability mass outside the manifold, while constraint-aligned sampling concentrates fully on valid residue lanes.

Thus, sampling determines access to structure.

4 Representation and Recovery

4.1 NMF Recovery

Nonnegative matrix factorization (NMF) recovers the residue manifold compactly:

$$X \approx WH,$$

where H aligns with residue lanes.

4.2 SAE Dilution

Sparse autoencoders (SAE) may introduce:

- redundant features
- inactive features
- diffuse feature support

Thus, representation determines whether structure is preserved or degraded.

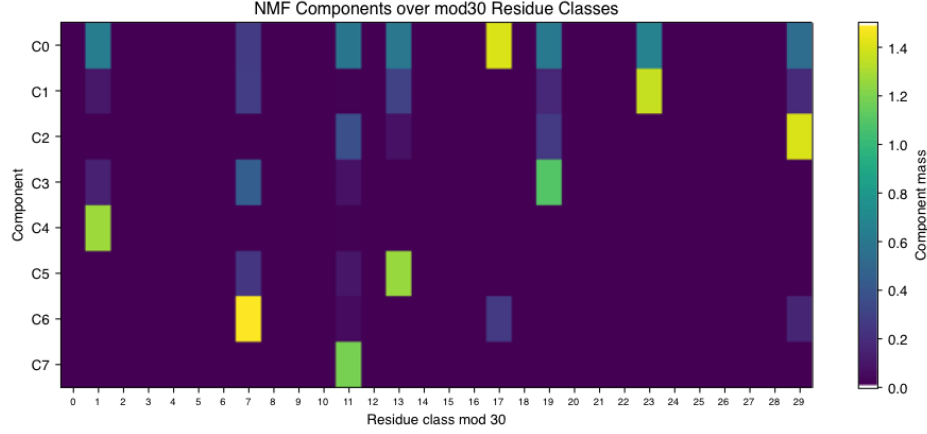


Figure 3: **Compact recovery.** NMF components align directly with residue lanes, demonstrating efficient and structured representation.

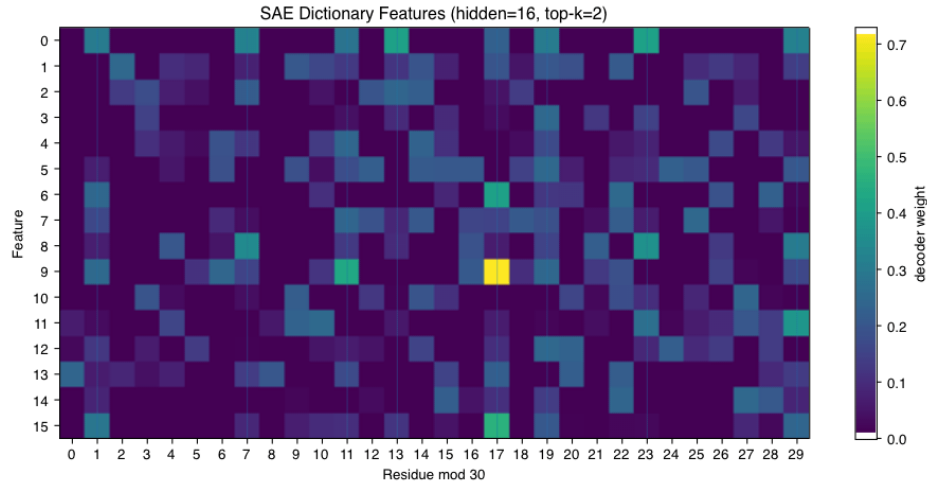


Figure 4: **Structural dilution.** SAE features distribute across residues, producing redundancy and reduced alignment with the underlying manifold.

5 Phase Diagram of Structural Regimes

We combine results into a phase diagram over model capacity.

Distinct regimes emerge:

- Recovered (aligned, compact)
- Partial
- Diluted
- Fragmented

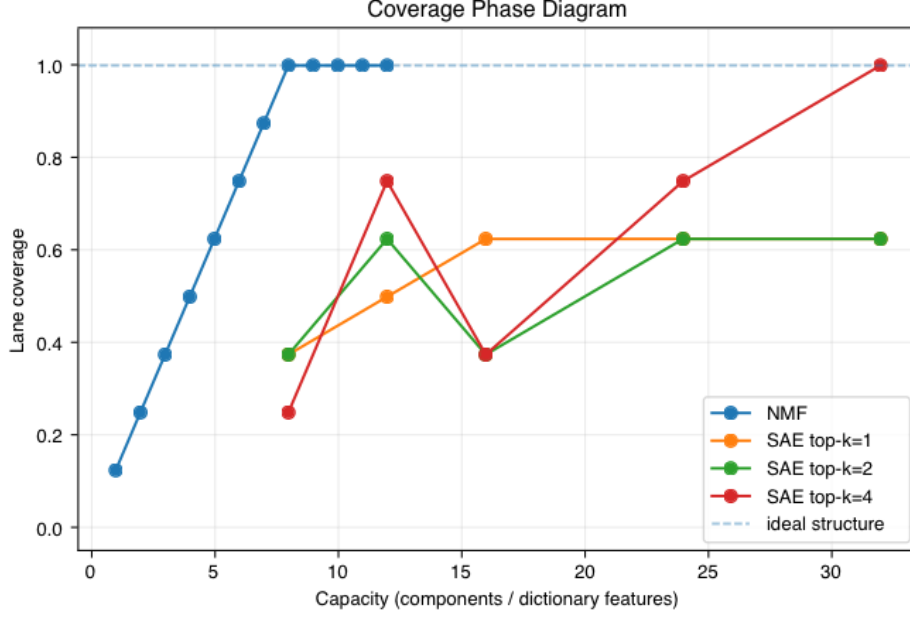


Figure 5: **Representation regimes.** NMF achieves full coverage at low capacity, while SAE exhibits partial, fragmented, and diluted regimes as capacity increases.

6 Constraint-Guided Coherence Score (CGCS)

We define:

$$\text{CGCS} = (\text{coverage})(\text{alignment})(\text{redundancy penalty})(\text{dead feature penalty})(\text{reconstruction penalty}).$$

CGCS is bounded in $[0, 1]$.

We define a phase-lock threshold:

$$\text{CGCS} \geq \frac{24}{25}.$$

This provides a quantitative measure of structural quality.

7 Phase-Lock Geometry

We interpret CGCS geometrically using cosine similarity:

$$\cos \theta = \frac{A \cdot B}{\|A\| \|B\|}.$$

Structure quality corresponds to alignment angle θ :

- $\theta \approx 0^\circ$ — fully aligned
- $\theta \leq 45^\circ$ — phase-locked
- $\theta > 45^\circ$ — degraded

This connects representation quality to geometric alignment.

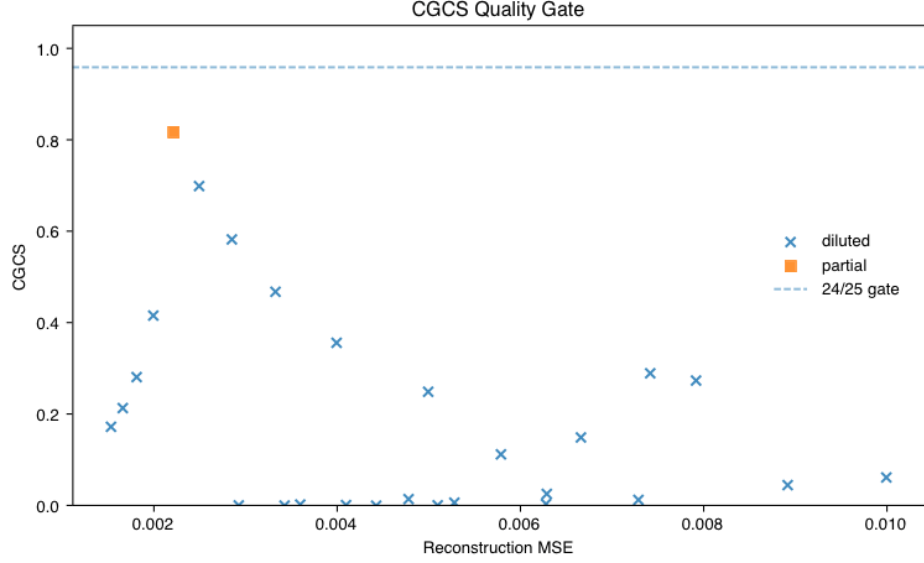


Figure 6: **Phase-lock threshold.** CGCS separates phase-locked representations from degraded ones, showing that reconstruction accuracy alone does not determine structural fidelity.

8 Discussion

We show that:

- Structure exists independently of representation
- Sampling determines access to structure
- Representation determines preservation of structure

Compact representations preserve structure, while overparameterized sparse representations may degrade it.

9 Conclusion

Residue-manifold learning provides a minimal setting to study structural fidelity. CGCS unifies coverage, alignment, and representation quality into a single metric. Its geometric interpretation reveals a phase-lock boundary governing structure preservation.

References

- [1] E. Tang, *A quantum-inspired classical algorithm for recommendation systems*, arXiv:1807.04271 (2018).

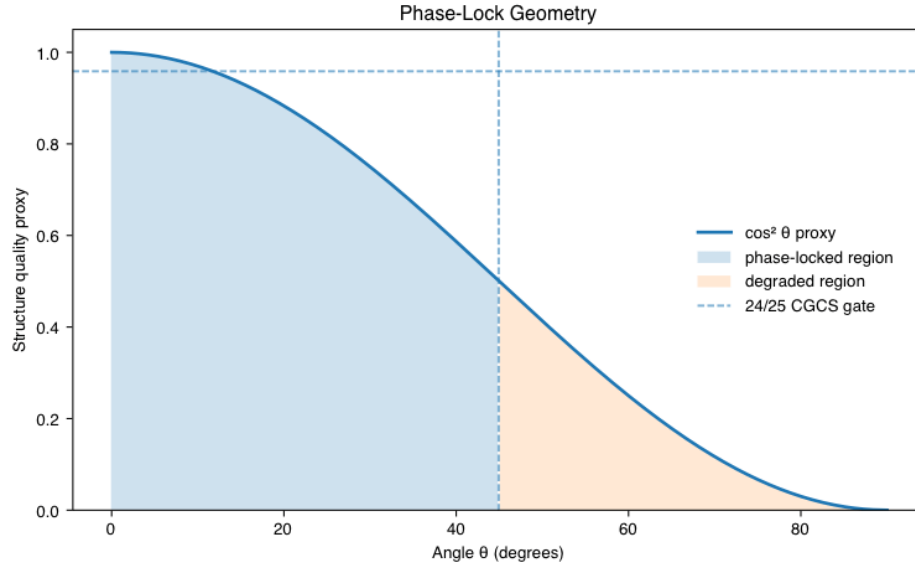


Figure 7: **Geometric interpretation.** Structural fidelity corresponds to alignment within a cosine-constrained phase space. The 45° boundary defines a phase-lock region consistent with the CGCS threshold.